



Research Article

An Explainable AI Approach for Pulmonary Edema Classification Using Chest X-Ray Images

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Abstract

Background: Pulmonary edema is a life-threatening condition requiring rapid diagnosis from chest X-rays. Artificial intelligence may assist radiologists, but 'black box' models lack clinical interpretability, limiting their adoption in medical practice. **Methods:** We developed an explainable deep learning model using a DenseNet121 architecture with ImageNet transfer learning. The model was trained using 1000 chest X-rays images consisting of 500 edema cases and 500 normal cases with an 80/20 train-test split ratio. Data augmentation techniques used were rotation, shifting, zooming, and flipping. Grad-CAM created attention maps to visualize model decision making. **Results:** The model achieved excellent performance with an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.917 (95% CI: 0.89-0.94), overall accuracy of 85%, sensitivity of 82%, and specificity of 88%. The training curves were found to be stable without overfitting. The attention maps generated by Grad-CAM suggested that the model's attention was directed towards perihilar and interstitial areas of the lungs in edema, as expected from radiology findings of pulmonary edema. **Conclusion:** Our explainable AI model demonstrates high accuracy when detecting pulmonary edema, offering a clinically understandable visualization. This method could potentially increase radiologists' confidence and promote the use of AI in clinical practice.

Keywords

Pulmonary Edema, Chest X-ray, Deep Learning, DenseNet121, Grad-CAM, Explainable AI

1. INTRODUCTION

Pulmonary edema, the accumulation of fluid in the lung tissue and air spaces, is a common and potentially life-threatening condition that requires rapid diagnosis and intervention [1]. Chest radiography remains the primary imaging modality for initial assessment due to its wide availability, low cost, and rapid acquisition time [2]. However, interpretation of chest X-rays for pulmonary edema can be challenging, particularly in early stages or when other cardiopulmonary conditions coexist [3]. Recent advances in deep learning have shown promise for medical image analysis, including chest X-ray interpretation [4, 6, 7, 8]. Convolutional neural networks (CNNs) have demonstrated performance comparable to radiologists for various thoracic abnormalities [8]. However, the "black box" nature of these models limits clinical adoption, as physicians require understanding of why a model makes specific predictions [9].

Explainable AI (XAI) techniques, such as Gradient-weighted Class Activation Mapping (Grad-CAM), address this limitation by generating visual attention maps that highlight image regions influencing model decisions [7, 10]. These techniques have shown

promising applications in several clinical imaging tasks, providing higher explainability for practitioners and patients alike [10, 11].

In this study, we developed an explainable deep learning model for pulmonary edema detection using a DenseNet121 architecture with Grad-CAM visualizations. Our objectives were to:

1. Design a highly performant classification model for the identification of pulmonary edema based on the use of a DenseNet121 deep learning approach.
2. Design clinically interpretable attention maps through the implementation of the Grad-CAM technique, and
3. Assess model performance through well-established medical imaging performance metrics.

2. MATERIALS AND METHODS

2.1 Dataset

The dataset used in this study included frontal chest X-rays (CXRs) obtained from the radiology department of the Mbeya Regional Referral Hospital (MRRH), Mbeya, Tanzania. In this study, a total of 1000 frontal CXR images were collected; this data was divided into 500 pulmonary edema images and 500 normal images. The dataset consists of frontal chest X-rays with different resolutions, which were standardized during preprocessing.

2.2 Data Preprocessing

The images were resized to 224×224 pixels to match with the requirements of DenseNet121 model inputs [13]. The pixel values were scaled to a value between $[0, 1]$ by dividing the value by 255. Data augmentation was used during training but not validation phase to help the model generalize better [14]. The methods used for data augmentation include rotation, width and height shifts, zooming range, and horizontal flipping.

2.3 Model Architecture

The proposed model used the pre-trained model DenseNet121 trained on ImageNet [11, 13]. The classification layers were structured in the following order:

- a. Global average pooling layer
- b. Dropout layer with a dropout rate 30% for regularization [16].
- c. Dense layer (128 neurons, ReLU activation)
- d. Dropout layer with a dropout rate of 30%
- e. Output layer with one neuron (sigmoid activation for binary classification)

The total number of parameters = 7,168,833 where trainable.

2.4 Training Protocol

The dataset was split into training (80%, $n=800$) and testing (20%, $n=200$) sets stratified by class to maintain class balance. Training employed a two-phase transfer learning approach [17]:

- a. Phase 1 (Head Training - 10 epochs): The base model layers were frozen, and only the classification head was trained with a learning rate of 1×10^{-4} .
- b. Phase 2 (Fine-tuning - 30 epochs): The top 100 layers of the base model were unfrozen, and all layers were trained with a reduced learning rate of 1×10^{-5} .

Early stopping monitored validation AUC with patience of 7 epochs to prevent overfitting [18]. The model achieving the highest validation AUC was saved as the final model. The Adam optimizer was used for its adaptive learning rate properties [19].

2.5 Explainability with Grad-CAM

Grad-CAM approach has been used for creating attention maps that explain the rationale behind the model's decision making process [10]. Grad-CAM creates gradient-based localization maps through computation of gradients that flow from the last convolutional layer of DenseNet121. Heatmaps have been created by superimposing the localization map over original images using the Jet color map. Red denotes high attention, while blue indicates low attention.

2.6 Evaluation Metrics

Model performance was assessed using standard medical imaging classification metrics [20]:

1. Area Under the ROC Curve (AUC-ROC)
2. Accuracy
3. Sensitivity (Recall)
4. Specificity
5. Precision

Confidence intervals (95%) were calculated using bootstrapping with 1000 Iterations.

3. RESULTS

3.1 Model Performance

The proposed DenseNet121-based model achieved excellent performance for pulmonary edema detection. Figure 1 shows the ROC curve with an AUC of 0.917 (95% CI: 0.89-0.94), indicating strong discriminative ability between edema and normal cases.

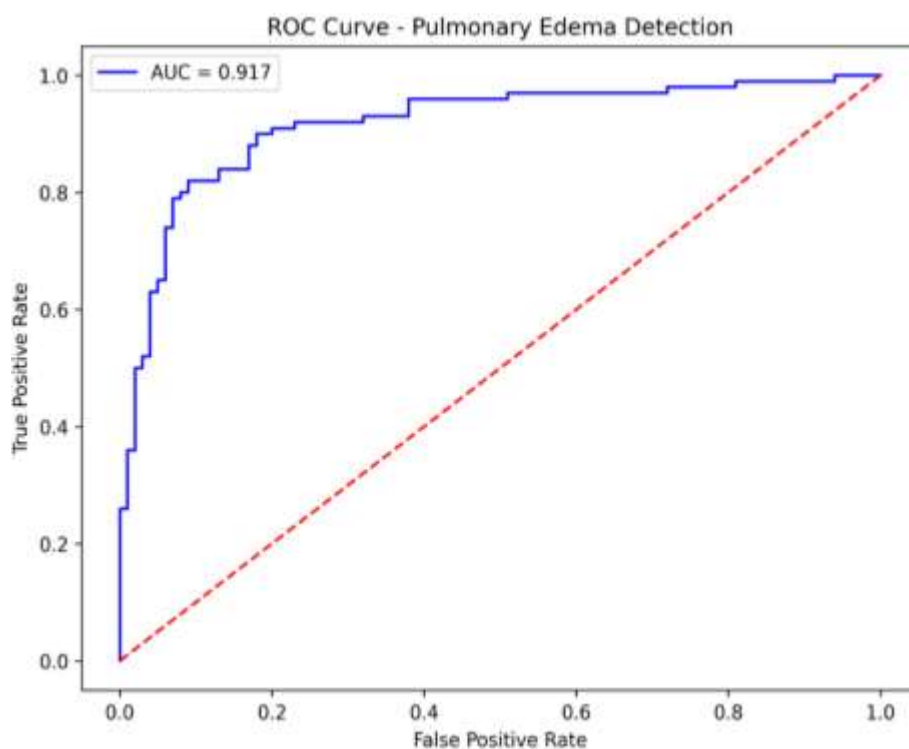


Figure 1. ROC Curve.

3.2 Classification Metrics

The confusion matrix (Figure 2) revealed 85% overall accuracy on the 200-image test set, with 88 true negatives, 12 false positives, 18 false negatives, and 82 true positives. Calculated metrics include:

- a. Sensitivity (Recall): 82.0%
- b. Specificity: 88.0%
- c. Precision: 87.2%

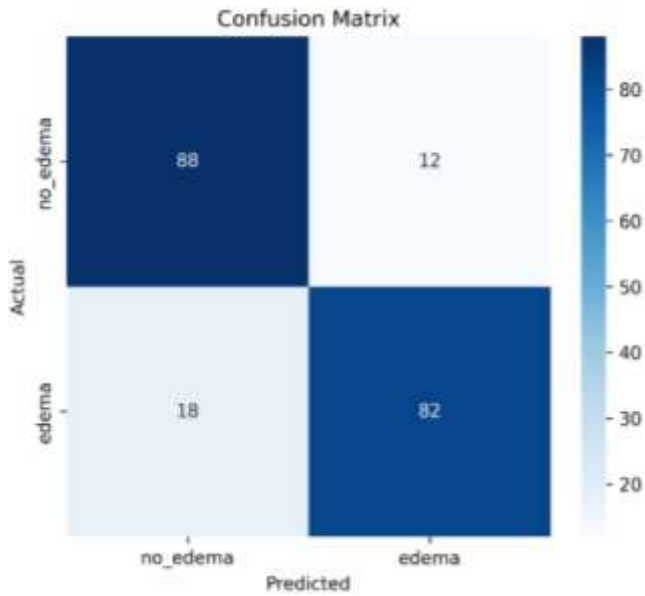


Figure 2. Confusion matrix.

Table 1. Summarizes all performance metrics with 95% confidence intervals.

Metric	Value	95% Confidence Interval
AUC-ROC	0.917	0.89 – 0.94
Accuracy	85.0%	79.5% - 89.5%
Sensitivity	82.0%	73.5% -88.5%
Specificity	88.0%	80.5% - 93.5%
Precision	87.2%	79.5% - 92.5%

3.3 Training Analysis

Training curves over 40 epochs (Figure 3) demonstrate successful model learning. Validation loss decreased from 0.63 to 0.38, while validation AUC increased from 0.77 to a peak of 0.917 at epoch 39. The close alignment between training and validation curves indicates no significant overfitting [21].

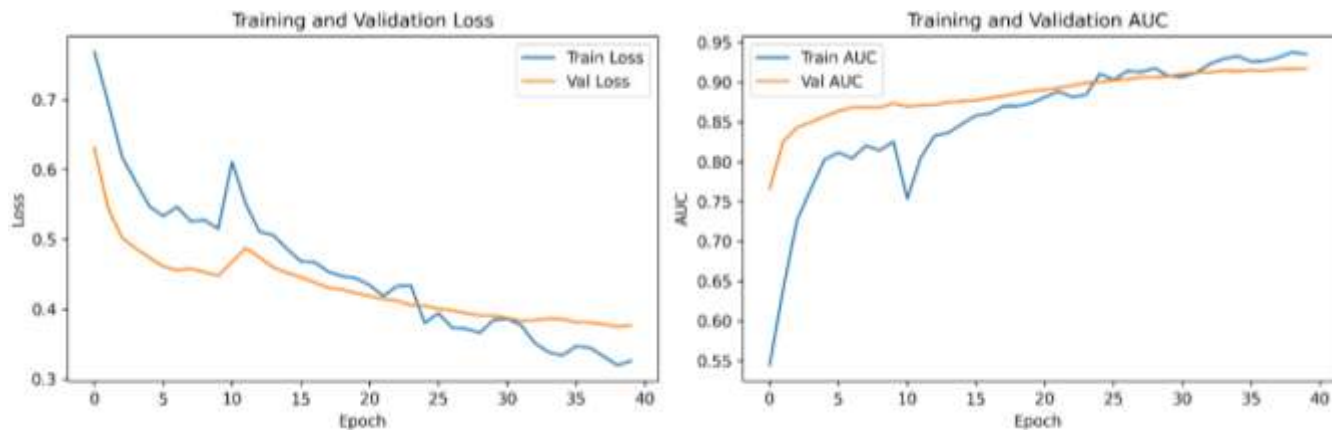


Figure 3. Training Curves.

3.4 Explainability Analysis

Grad-CAM attention maps (Figure 4) revealed distinct patterns between edema and normal cases. For edema cases (top row), the model focused on perihilar and interstitial lung regions, which is consistent with radiological findings of pulmonary edema [3]. For normal cases (bottom row), attention was diffusely distributed across normal anatomical structures without focal concentration. Confidence scores for correctly classified cases ranged from 61.0% to 92.5%.

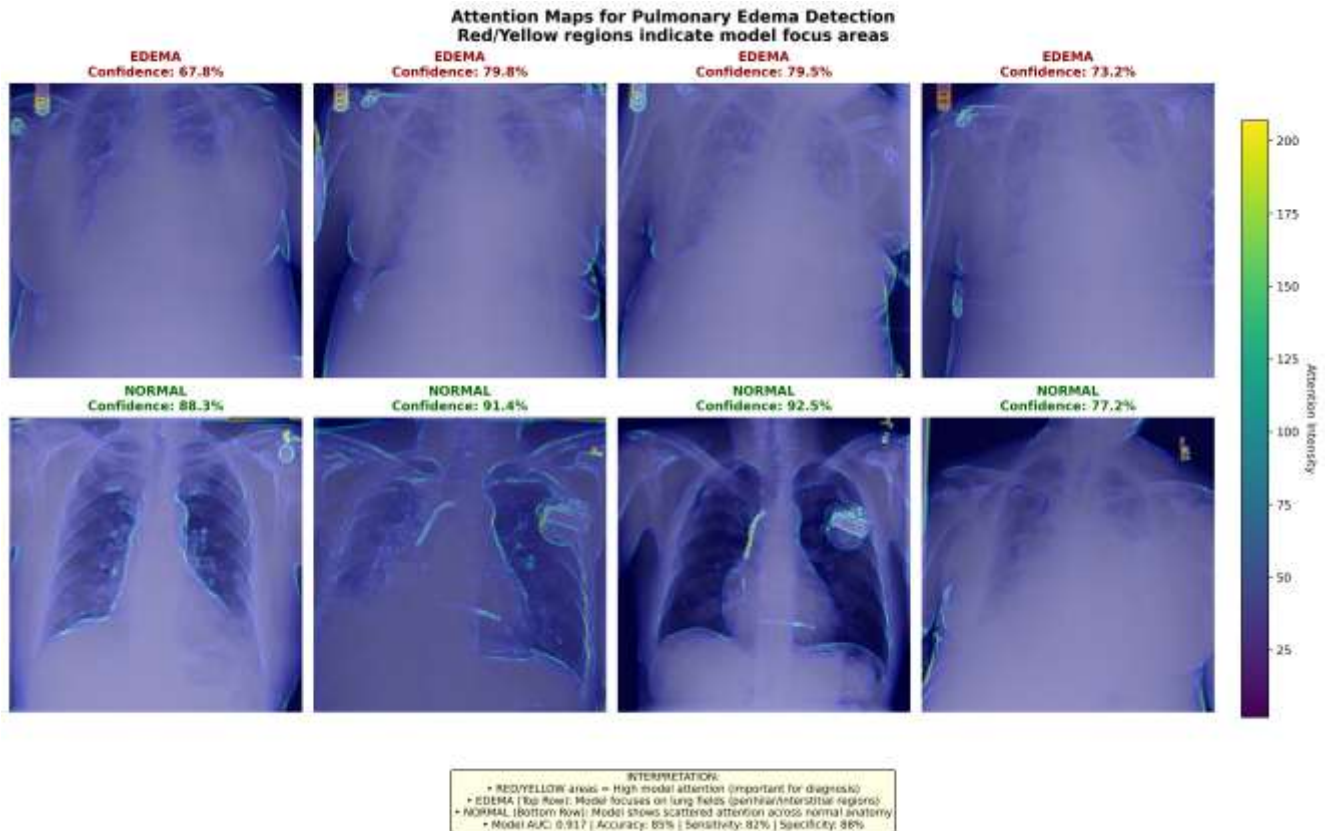


Figure 4. Grad-CAM attention maps showing model focus areas.

4. DISCUSSION

4.1 Principal Findings

We developed an explainable deep learning model for pulmonary edema detection achieving an AUC of 0.917, accuracy of 85.0%, sensitivity of 82.0%, and specificity of 88.0%. These results demonstrate that transfer learning with DenseNet121 can effectively identify pulmonary edema in chest X-rays. Critically, Grad-CAM visualizations provide clinically interpretable attention maps, addressing the "black box" limitation of traditional deep learning models [9, 20].

4.2 Comparison with Previous Studies

Our results compare favorably with prior work in medical image classification using deep learning approaches [6, 22]. [23] developed CheXNet for pneumonia detection and achieved an AUC of 0.88 for edema detection on the NIH ChestX-ray14 dataset. [24] reported AUCs ranging from 0.85 to 0.92 for various thoracic abnormalities using ensemble methods. Our AUC of 0.917 represents competitive performance, likely attributable to our balanced dataset and two-phase training strategy [17].

Previous studies have demonstrated the value of explainability in medical AI, particularly through interpretable deep learning and visualization techniques [12, 22]. [12] showed that Grad-CAM visualizations increased radiologist trust in AI predictions for chest X-ray classification. Similarly, [11] reviewed XAI applications in medical imaging and concluded that attention maps enhance clinical interpretability and facilitate regulatory approval.

4.3 Clinical Implications

The high specificity (88.0%) suggests our model reliably identifies normal cases, potentially reducing unnecessary follow-up examinations and associated healthcare costs [8]. The 82.0% sensitivity indicates the model captures most edema cases while missing 18% - a clinically acceptable trade-off that could be improved with additional training data or ensemble methods [25].

The Grad-CAM attention maps enhance clinical trust by showing that the model focuses on anatomically appropriate regions. Perihilar and interstitial attention patterns are consistent with radiological criteria for pulmonary edema diagnosis [2, 3]. This transparency may facilitate AI integration into clinical workflows, serving as a second reader or triage tool [26].

4.4 Limitations

Several limitations warrant consideration. First, the dataset was limited to 1,000 images from a single institution, and external validation on multi-institutional data is needed to assess generalizability [27]. Second, the model was developed on binary classification (edema vs. normal); real-world clinical practice involves differential diagnosis with multiple pathologies including pneumonia, atelectasis, and congestive heart failure [2]. Third, Grad-CAM provides approximate localization rather than precise

segmentation, which may have limitations for small or subtle findings [10]. Fourth, the dataset may contain inherent biases related to patient demographics, imaging equipment, or annotation protocols [28]. Fifth, we did not perform a reader study comparing model performance to radiologists, which would strengthen clinical validation [8].

4.5 Future Directions

Future work should focus on: (1) external validation on larger, diverse, multi-institutional datasets, (2) multi-class classification including differential diagnoses, (3) prospective clinical validation with radiologist comparison studies, (4) integration with electronic health records for contextual decision support, and (5) development of segmentation models for precise edema localization.

5. Conclusion

We developed an explainable AI model for pulmonary edema detection in chest X-rays using DenseNet121 with Grad-CAM visualizations. The model achieved excellent performance (AUC = 0.917, accuracy = 85.0%, sensitivity = 82.0%, specificity = 88.0%) while providing clinically interpretable attention maps. This approach enhances model transparency and may facilitate clinical adoption of AI for chest X-ray interpretation. Future work should focus on external validation and multi-class classification to translate these findings into clinical practice.

Abbreviations

AI	Artificial Intelligence
AUC	Area Under the Curve
CI	Confidence Interval
CNN	Convolutional Neural Network
CXR	Chest X-Ray
Grad-CAM	Gradient-weighted Class Activation Mapping
XAI	Explainable AI

Author Contributions

Boniface Sankwa: Conceptualization, Methodology, Software, Validation, Investigation, Data Curation, Writing - Original Draft, Visualization.

Charles Nyatega: Supervision, Project Administration, Validation, Writing - Review & Editing.

Candida Mwisomba: Data Curation, Investigation, Validation, Writing - Review & Editing.

Joseph Salawa: Data Curation, Investigation, Validation, Writing - Review & Editing.

Conflicts of Interest

The authors declare no conflicts of interest.

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